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Paleomagnetic dating of continental geological formations: Strong diachronism evidenced in the Saharan platform and geodynamical implications

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ABSTRACT

The paleomagnetism is a powerful tool to date formations that have age not constrained by paleontological, stratigraphical or radiochronological data. It was applied, on the western border of the Murzuq basin in Algeria (Saharan platform), to the Zarzaïtine formation, attributed to a Middle–Upper Triassic–Lower Jurassic age. Comparison of the obtained paleomagnetic pole with previous poles from the same geological formation outcropping in another basin and from other Carboniferous to Lower Mesozoic African formations yielded a clearly older age (Late Permian) than expected. That evidences a strong diachronism (at least 40 My) of the deposition of this formation on the Saharan platform. The post-Hercynian structural evolution was therefore different according to the parts of this platform, with significant differential vertical tectonic movements. The latter were at the origin of erosion, hiatus or sediments deposition according to areas.

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1. Introduction

The paleomagnetic tool is mainly used to determine movements of the main tectonic plates and paleocontinental reconstruction (e.g. Van der Voo, 1988, 1990; Torsvick et al., 2012), or to determine more or less local tectonic rotations (e.g. in the Western Alps – Westphal, 1973; Henry, 1992; Thomas et al., 1999; Maffione et al., 2008). It is also applied for relative or absolute dating in sedimentary or volcanic sequences with the magnetostratigraphic approach that is mainly based on the polarity of the paleomagnetic direction (e.g. Zhu et al., 2003; Hüsing et al., 2011). However, by comparison of this direction with a previously established reference curve (variation of the direction, measured in well-dated "objects", as a function of time), it could be as strong tool for dating. Presently, such application is widely used for dating within a recent period for archeological structures (Thellier, 1938; Le Goff et al., 2002) or lava flows (Tanguy et al., 2007). For that, a reference curve, based on declination and inclination of the measured direction on dated recent "objects" was determined and is still improved in different countries (e.g., Thellier, 1981; Bucur, 1994; Gallet et al., 2002). This separation by areas is made in order to have very precise curve independent of regional anomalies of the Earth magnetic field, which does significantly change in such short period.

2. Paleomagnetic dating

Reference curves for paleomagnetic data are built with another approach (Creer et al., 1954), to be independent of the location of the studied object in very large zones. It is based on paleomagnetic data from well-dated geological formations. An Apparent Polar Wander Path (APWP) is a plot of the sequential positions of paleomagnetic poles from a particular area (tectonic plate, continent, etc.), usually shown on the present geographic grid (see Butler, 1992). The constraining factor to determine such areas is the lack of local rotations within them. It is however sometimes possible to correct such rotations, like for Lachland belt in Australia (McElhinny et al., 2003), or even to use paleocontinental reconstruction to have reference curve for very large areas (Besse and Courtillot, 2002; Torsvick et al., 2012).

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For geological "objects", paleomagnetic dating was mostly employed to determine the age of a geological event that yielded remagnetization of the studied rocks (e.g. Rouvier et al., 2001; Ricordel-Prognon et al., 2010; Lamali et al., 2013). Dating using primary magnetization has been rarely used for geological formations (e.g. Censier et al., 1995), except for ore deposits (e.g. Leach and Sangster, 1993; Symons and Sangster, 1994).

Among geological formations, continental deposits are often those with the poorest dating criteria, and they can represent very large window of time, particularly in cratonic areas. The Permian period in the Saharan platform is a typical example of such "geological gap". The last marine well-dated Paleozoic levels are of Upper Carboniferous (Moscovian) age. The age of the overlying continental levels is very imprecisely defined. These levels are in part of the platform of Stephano-Autunian age, and are themselves covered by continental deposits, called "Zarzaïtine" formations, locally dated of the Middle–Upper Triassic–Lower Jurassic times. In other areas, these last deposits are directly in contact with the marine Moscovian formation. To better understand this "geological gap" in the Saharan platform, we therefore applied a paleomagnetic dating to compare the age of the "Zarzaïtine" formation in different areas (Fig. 1).

The interest of such a dating was not some local precision in stratigraphy, but concerns the effects of post-Hercynian structural evolution in these areas very far from the continental collision suture between Gondwana and Laurasia in Western Africa. Post-Hercynian tectonic events have been already pointed out on basins of northwestern border of the Hoggar (Smith et al., 2006; Derder et al., 2009). They represent key information for petroleum research. In Niger, in "Monts Doumergue" area about 60 km south of the Algerian boundary, a cartographic discordance shows Zarzaïtine series directly overlying the Lower Carboniferous formations and even Cretaceous levels overlying these formations (Fig. 1d). Unfortunately, this area, very isolated in the northern Niger, is presently not accessible for safety reasons and we tried to obtain new data on the Zarzaïtine formation, to compare them with previous results obtained in the Illizi basin (Kies et al., 1995). To this aim, the Anaï area in Algeria was selected. No significant post-Hercynian local rotations affected this part of the Saharan platform. The new studied sites correspond to the western border of the Murzuq basin.

3. The Zarzaïtine series

The Zarzaïtine formation represents continental deposits largely developed in South Algeria, Libya and Niger. The underlying «Tiguentourine» continental formation is considered as missing in a large part of these countries, being for example restricted in Libya to some deep basins (Hallett, 2002). Due to bad outcrop conditions, it is sometimes not possible to determine what formation, Tiguentourine or Zarzaïtine, overlays the last marine Carboniferous well-dated levels. In areas without paleontological arguments the actual age of the Zarzaïtine deposits is in practice unknown. This is

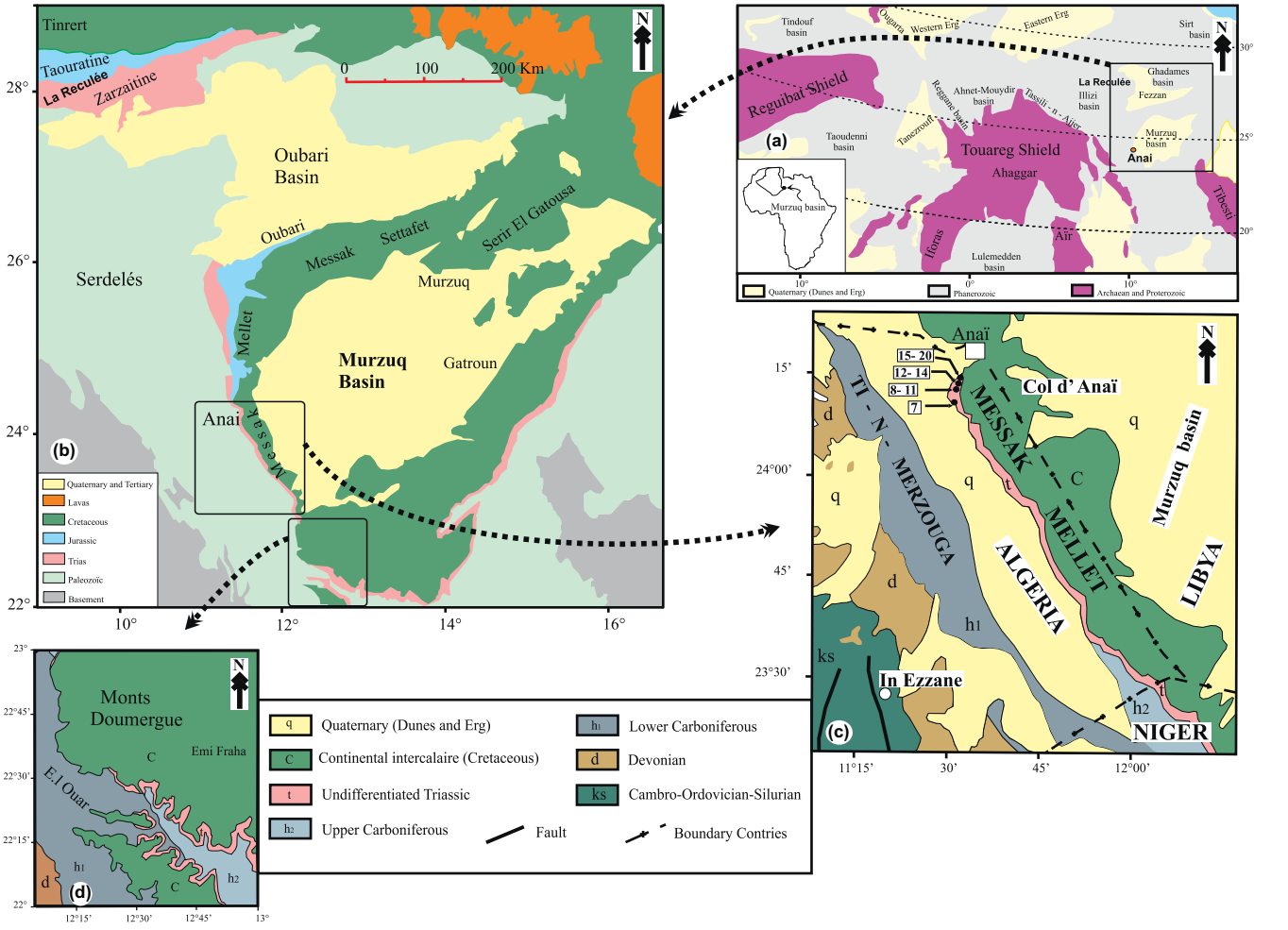


Fig. 1. Map of the studied area (after the "Carte Géologique du Nord-Ouest de l'Afrique – Sahara Central", 1962). Location of the sampling sites on figure c.

particularly the case on the southwestern border of the Murzuq basin in the borders of Libya, Algeria and Niger (Fig. 1a–c).

3.1. The Upper Carboniferous–Lower Mesozoic section in the Illizi basin

The reference section for this period is located in the Illizi basin in Algeria, at "La Reculée" area (Fig. 2). Overlying the El Adeb Larache Moscovian marine formation, the lower Tiguentourine series is assumed to be of Stephano-Autunian age (Attar et al., 1981). The upper Tiguentourine formation remains undated. Tiguentourine rocks comprise mainly shales, muddy siltstones and sandstones. The Hercynian unconformity at their top (Nedjari et al., 2009) marks the end of the Paleozoic formations. The overlying Zarzaïtine series (Fig. 2) is well-known in the In Amenas area since Lehman (1957) and Busson (1970) and attributed to Triassic and Jurassic ages. The lower Zarzaïtine formation comprises predominantly coarse white to red sandstones, with some interbedded white or red clay levels. Vertebrate fauna and in particular Stegocephalia, Sauria and fishes gave a Middle Triassic to Rhaetian age (Lehman, 1971; Bourquin et al., 2010; Aït Ouali et al., 2011). The upper parts of the Zarzaïtine formations are attributed to Lower–Middle Jurassic and are overlain by the Taouratine series of Upper Jurassic–Lower Cretaceous age (de Lapparent et al., 1958).

3.2. The Upper Carboniferous–Mesozoic formations in the Murzuq basin

In the Murzuq basin, uplift between 300 and 2000 m affected all the area after the Moscovian and the Tiguentourine formation

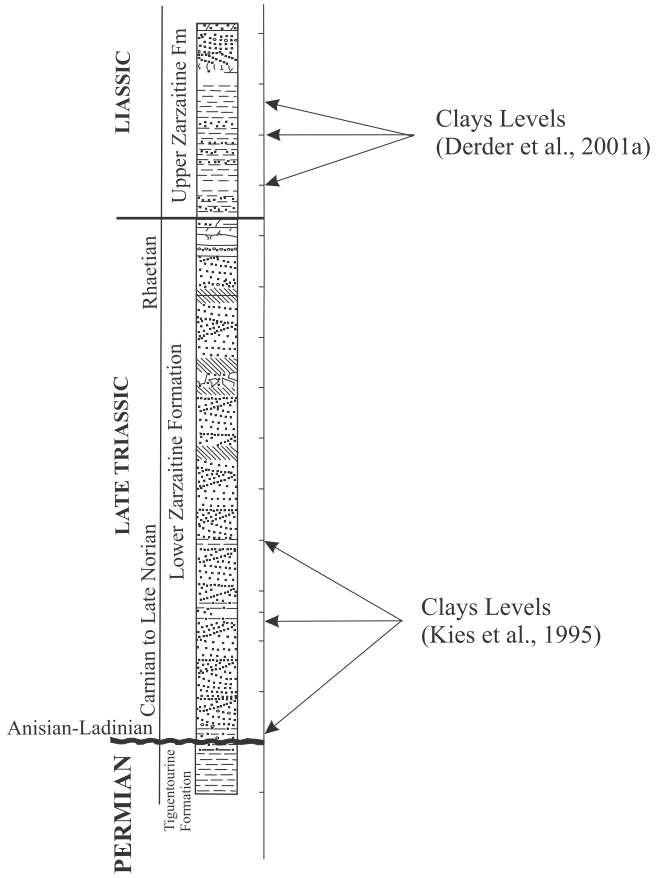


Fig. 2. Sedimentological log of the Zarzaïtine formation in La Reculée area (after Bourquin et al., 2010), with indication of the studied levels in previous paleomagnetic studies.

should be only present in newly-formed interior basins (Hallett, 2002). In addition, none Permian outcrops are known throughout Libya (Tawadros, 2000). In Algeria, the single Murzuq basin section, which should be equivalent to that of La Reculée, is located at Anaï area in the borders of Libya, Algeria and Niger (Fig. 1b). On the available geological maps (Fig. 1c; Freulon, 1964), the Paleozoic series are directly overlain by the Zarzaïtine formations, of assumed Triassic to Jurassic age by analogy with the Illizi basin (Freulon, 1964). Indeed, a Triassic to Jurassic age was confirmed for these formations on the eastern border of the Murzuq basin (Lejal-Nicol and Klitzsch, 1975).

4. Paleomagnetic analysis

4.1. Sampling and analysis procedure

Most of the western border of the Murzuq basin corresponds to a tabular area with a weak NNE dip. Outcrops conditions are very bad, the fallen rocks from the overlying Cretaceous cliff covering almost all the slope and the main valley. Only some red levels can be observed locally in deep secondary talwegs. These levels have facies (clays to fine-grained sandstones) very similar to that of the interbedded clays previously studied (Kies et al., 1995) within the Lower Zarzaïtine formation at La Reculée area, in the Illizi basin (Fig. 2). No fossils were found within the clays at Anaï area, except a small piece of wood. The bottom and top (up to the Cretaceous cliff) parts of the formation do not outcrop. 222 oriented cores, obtained with a gasoline-powered portable drill and oriented with magnetic and sun compasses, were collected in 14 sites along 4 sections (Fig. 1c). However, 23 of these cores were too fragile and lost during transport and specimens preparation. One of the sections corresponds to a single site located near the "Col d'Anaï" well and the three other sections are in intermediate locations between this well and the Anaï pass, each of the sites along a same section representing a different stratigraphic level. Each section corresponds to about 30 m of thickness of the stratigraphical series. Unfortunately, outcrop conditions made impossible a correlation between the sections. At the well area, the relatively important dip (35° compared to 4° in the other sites) could suggest a sliding of a very large block. Taking into account the general structural context, comparison of sites elevation however indicates that the sliding, if existing, should be weak.

One to three specimens were cut from each core in order to obtain additional specimens for pilot studies and rock magnetism analyses. Prior to any demagnetization analysis, the specimens were stored in a zero field for at least one month, in order to reduce possible viscous components. The Remanent Magnetization of the specimens was measured using a JR5 spinner magnetometer (AGICO, Brno). Several pilot specimens from each site were subjected to a stepwise alternating field (AF) demagnetization up to 100 mT or thermal demagnetization up to 670 °C in order to characterize the components of remanent magnetization.

The results obtained with the pilot studies showed that the AF did not allow complete demagnetization. Thermal demagnetization was then applied to the remaining specimens. It was performed using a cylindrical furnace (ASC Model TD48-SC) with a fast heating level (45 min to reach 600 °C) while forced air enhanced the cooling (20 min). In order to isolate correctly and identify the components of magnetization, numerous steps (between 20 and 26 according to samples) were performed with increments ranging from 100 °C at the lowest temperatures to 5 °C at the highest temperatures. Demagnetization data analysis was carried out using classical methods: the directions of the magnetization components were computed using principal component analysis

(Kirschvink, 1980) and plotted on orthogonal vector plots (As and Zijderveld, 1958; Zijderveld, 1967); the remaining vectors and vectorial differences of the magnetization were plotted on equal-area projections. The mean directions of the magnetization components were determined using Fisher (1953) statistics.

4.2. Rock magnetic study

The Natural Remanent Magnetization (NRM) intensity is generally weak but measurable with a JR5 spinner magnetometer (sensivity around 2×10^{-6} A/m), varying from 4×10^{-4} A/m to 3×10^{-2} A/m, with a mean of 9×10^{-3} A/m. Thermomagnetic curves (low-field magnetic susceptibility as a function of temperature) have been measured for representative samples by heating using the CSL equipment and the CS2-3 oven under argon atmosphere, combined with the Kappabridge KLY3 (AGICO, Brno). Most samples show reversible curves during heating and cooling (Fig. 3), suggesting that no important mineralogical alterations occurred for these samples during heating up to 680 °C. At heating temperatures about 450–500 °C, the magnetic susceptibility first shows a large increase (Hopkinson effect) before a strong decrease up to 650–680 °C. This behavior is typical for hematite. Hysteresis loops were obtained for small cores (cylindrical samples of about 3 cm^3) using a laboratory-made translation inductometer within an electromagnet capable of reaching 1.6 T. Saturation of magnetization is not reached (Fig. 4) and the coercive forces are relatively high (remanent coercive force Hcr between 0.08 and 0.35 T). This confirms that the principal carrier of the magnetization is hematite. The thermal demagnetization curves of the NRM show maximum unblocking temperatures around 670–675 °C for all samples, also confirming hematite as principal magnetic carrier (Fig. 5).

Differences observed on the curves (Figs. 3–5) for different specimens could be related to slight facies variation (difference of proportion of clay minerals).

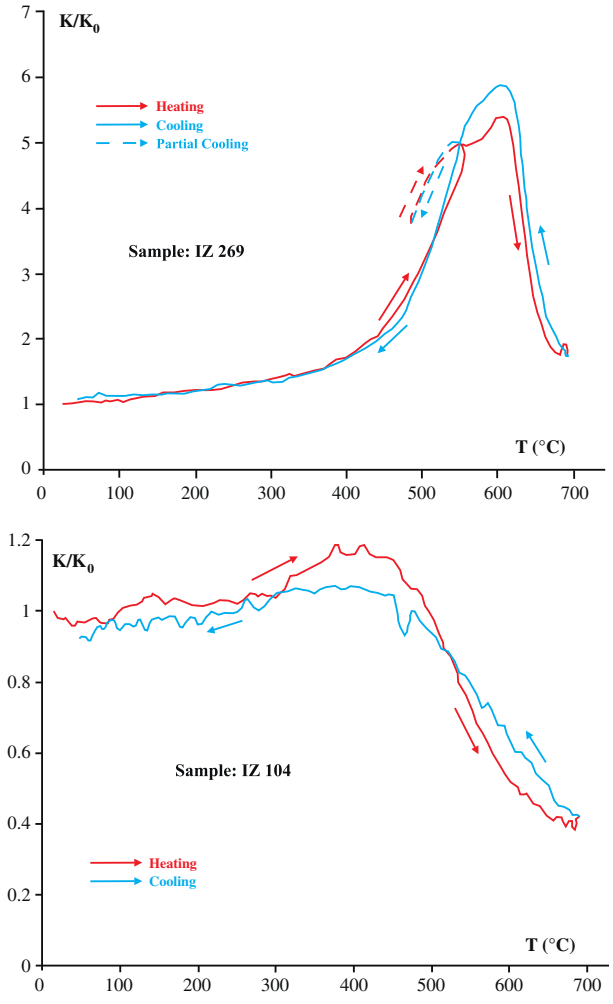


Fig. 3. Typical thermomagnetic curves: normalized low field susceptibility (K/K_o) as a function of the temperature T (in °C) for samples IZ269 and IZ104.

4.3. Paleomagnetic results

The NRM directions are distributed on stereographic projection on a girdle elongated between the present Earth magnetic dipole field direction and a SE direction. This display suggests a superimposition of at least two components in the NRM. During thermal demagnetization, the analysis of the NRM gives different kinds of evolution on the Zijderveld plot. Only 12 samples did not yield reliable results. Mostly a low temperature magnetization A is first eliminated. For part of the samples, two linear segments have been isolated at higher temperature, then corresponding to a magnetization B and a Characteristic Remanent Magnetization (ChRM) C (Fig. 6a and b). A ChRM is the stable magnetization component remaining after elimination of all the other components (it is characterized on Zijderveld diagram by straight lines converging towards the axes origin). In some cases, the first linear segment is missing or is replaced by a curved line and only the ChRM C can be isolated (Fig. 6c). In other cases, on the contrary, the ChRM C cannot be accurately defined and only the component B has been obtained. Ending, for 19 samples, no stable component was reached even for the highest temperatures (Fig. 6d) and only evolution on stereoplots toward the SE quadrant along a great circle (well-defined only for 9 samples) has been obtained.

The component A, generally isolated between the room temperatures and 300 °C, has in very few cases maximum unblocking temperatures beyond 300 °C and even sometimes reaching 670 °C (Fig. 6e), being then in fact a ChRM ($N = 15$ samples, $D = 359.5°$, $I = 39.2°$, $k = 32$, $\alpha_{95} = 6.0°$ and $D = 2.7°$, $I = 39.1°$, $k = 32$, $\alpha_{95} = 6.0°$, before and after dip correction, respectively). The magnetization A probably represents actually the superimposition of components of neighboring orientation, one of "viscous" origin A1 and another one A2 much more stable, sometimes giving even a ChRM.

The ChRM C is generally isolated, depending on samples, between 500–600 °C and 670–675 °C. In cases where the B component is missing, the threshold of the lowest temperatures of the component C could be as low as 300 °C. The ChRMs C are well clustered on the south-east quadrant with low inclination, for individual samples direction (Fig. 7a) as well as for mean sites directions (Fig. 7b). The mean direction of this ChRM C computed from 84 specimens has the following directions: $D = 143.2°$, $I = 8.1°$, $k = 19$, $\alpha_{95} = 3.7°$ and $D = 142.3°$, $I = 3.7°$, $k = 22$, $\alpha_{95} = 3.4°$, before and after dip correction, respectively. This result is similar to that obtained from the sites mean direction: $N = 11$ sites, $D = 143.3°$, $I = 9.2°$, $k = 35$, $\alpha_{95} = 7.8°$ and $D = 142.4°$, $I = 5.2°$, $k = 47$, $\alpha_{95} = 6.7°$. Using the combination of the ChRM C and the 9 well-defined great circles data (McFadden and McElhinny, 1988), the mean directions obtained for the different sites (Table 1) give almost the same results with one more site, that was previously discarded (only 2 ChRMs C): $N = 12$ sites, $D = 143.9°$, $I = 9.7°$, $k = 33$, $\alpha_{95} = 7.1°$ and $D = 143.0°$, $I = 5.8°$, $k = 42$, $\alpha_{95} = 6.2°$. The dispersion is smaller in stratigraphic coordinates than in geographic ones, but the fold test (McElhinny, 1964) is inconclusive.

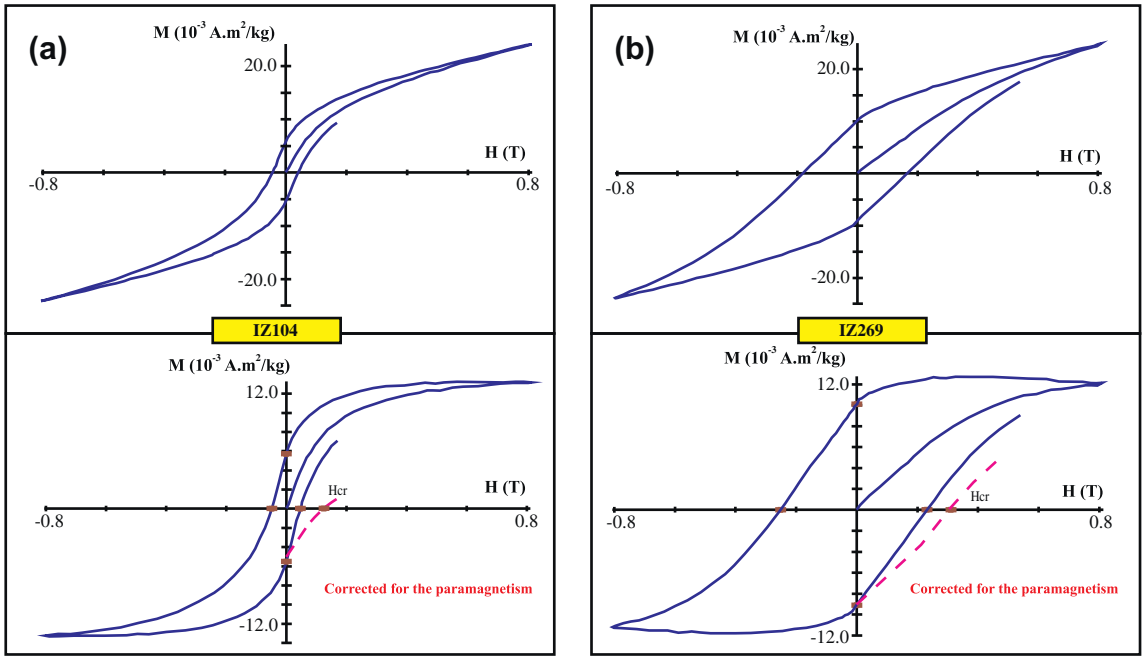


Fig. 4. Typical hysteresis loops before and after correction for paramagnetism: sample IZ104 (a) and IZ269 (b). Hcr is the remanent coercive force, field *H* in Tesla.

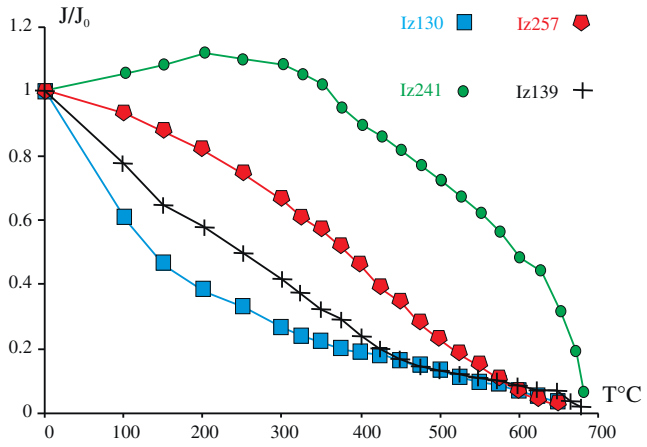


Fig. 5. Thermal demagnetization curves for samples IZ130, IZ241, IZ257 and IZ139.

5. Discussion

5.1. New paleomagnetic data

The paleomagnetic direction of the ChRM A2 is close to the "recent" Earth's magnetic field one. The associated paleomagnetic pole in geographic coordinates ($\lambda = 86.3°$S, $\phi = 191.5°$E) is far from African Lower Mesozoic poles, but close to the Cenozoic ones. Similar observation was made for different Paleozoic to Mesozoic formations, evidencing a regional remagnetization event during the post-Mesozoic period (Henry et al., 2004; Lamali et al., 2014). The ChRM A2 is thus interpreted as a chemical magnetic overprint acquired during relatively recent times.

The ChRM C data are homogeneous, but their directions show slight variations, in declination as well as in inclination (Table 1), according to the stratigraphic levels. This indicates that this ChRM likely does not correspond to a regional magnetic overprint, which should have given a same direction in all the levels. In addition, in the same area, a primary magnetization, constrained in age by a fold test, has been evidenced in a Moscovian formation (Amenna et al., 2014). For one hand, the single regional remagnetization events known in the Saharan platform are of Cenozoic (here, the component A2) and Permian ages and cannot be at the origin of the ChRM C. For another hand, the obtained directions are from fine-grained specimens without visible weathering. The studied formation rocks belong to a stable area. The slightly better clustering after bedding correction is however not a fully reliable indication for a dating relatively to folding because the single site (site 7) from an area with a significant dip belongs to a very large block suspected of sliding (it can be noticed that this site 7 should not introduce a bias, because the mean direction of the ChRM C without this site is not significantly different: $N = 11$ sites, $D = 141.3°$, $I = 8.0°$, $k = 40$, $\alpha_{95} = 7.3°$ and $D = 141.0°$, $I = 5.4°$, $k = 40$, $\alpha_{95} = 7.3°$ before and after dip correction, respectively). All these converging arguments indicate that the ChRM C is the primary magnetization, with therefore a corresponding paleomagnetic pole determined from data after bedding correction ($\lambda = 43.8°$S, $\phi = 70.9°$E, $K = 80$ and $A_{95} = 4.5°$).

On stereographic projection, the individual directions of the component B are very scattered. Depending of samples, this direction is either close to that of the ChRM A2 or close to that of the ChRM C, or between these two directions. The B component is thus interpreted as a composite magnetization resulting from the superimposition of a partial remagnetization A2 with the primary magnetization C. Such magnetic behavior has been already pointed out previously from other geological formations of Saharan platform (Henry et al., 1999; Merabet et al., 1999; Bouabdallah et al., 2003), and the component B will not further be taken it into account.

5.2. Comparison of the ChRM C with the previous data from the Illizi basin

Very few paleomagnetic data for a similar age have been obtained in Libya and Tunisia and in areas far from the Murzuq basin (Muttoni et al., 2001; Ghorabi and Henry, 1991) and comparison has to be made with the Illizi basin in Algeria where previous paleomagnetic studies have been carried out in the Lower

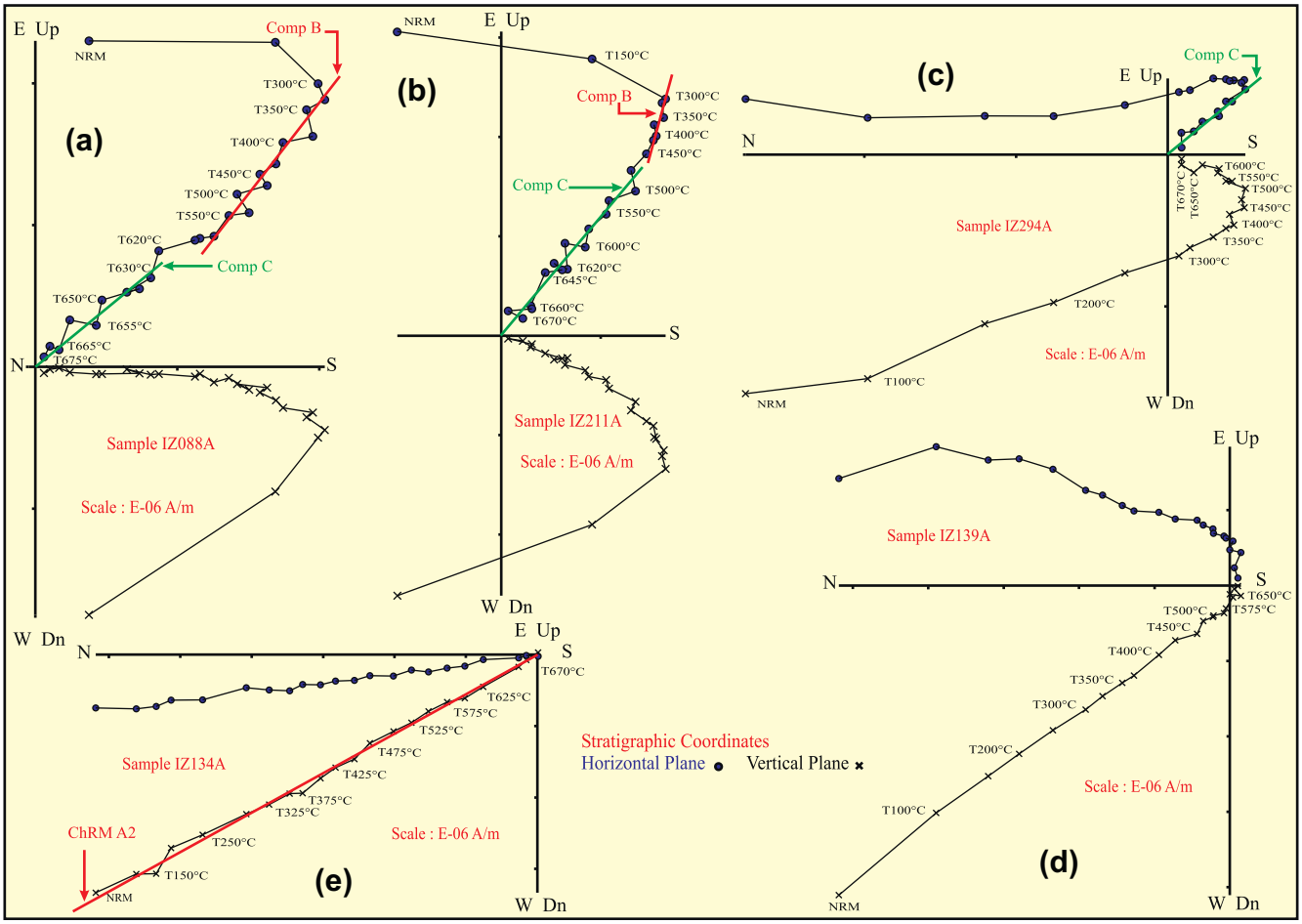


Fig. 6. Orthogonal vector plots (filled circles: horizontal plane, crosses: vertical plane), in stratigraphic coordinates for samples IZ088A (a), IZ211A (b), IZ294A (c), IZ139A (d) and IZ134A (e).

Tiguentourine (Derder et al., 1994) and Lower (Kies et al., 1995) and Middle (Derder et al., 2001a) Zarzaïtine formations. Though with very similar geological facies, the Lower Zarzaïtine at La Reculée and the Zarzaïtine at Anaï have clearly different magnetic characteristics. The maximum unblocking temperatures evidenced during thermal demagnetization are relatively low (less than 580 °C) at La Reculée and high (~670 °C) at Anaï, though a similar dominant main magnetic carrier (Hematite). ChRMs have normal and reversed polarities at La Reculée but only reversed polarity at Anaï. The Anaï paleomagnetic pole is in an intermediate position between the Stephano-Autunian Tiguentourine pole (Derder et al., 1994) and the Triassic–Lower Jurassic Zarzaïtine poles (Kies et al., 1995; Derder et al., 2001a) of the Illizi basin (Fig. 8; Table 2). It is therefore clear that the studied formation at Anaï cannot be contemporaneous of the Lower Zarzaïtine of La Reculée. The Fig. 8 also illustrates the location of the Anaï pole relatively to all the Illizi and Murzuq Upper Carboniferous–Lower Mesozoic ones. A large gap exists between the Upper Carboniferous–Lower Permian and Lower Mesozoic poles, and the new Anaï pole is precisely located within this gap.

5.3. Paleomagnetic dating

McElhinny et al. (2003) proposed a selection of paleomagnetic poles for the Gondwana (poles location for continents different from Africa was recalculated relative to the present Africa position according to the paleoreconstruction of the Gondwana at different periods). This selection was updated by Derder et al. (2006), Torsvick et al. (2012) and Amenna et al. (2014). Combined with data from Besse and Courtillot (2002) and Domeier et al. (2012), an improved APWP for Gondwana is presented on Fig. 8.

The paleomagnetic pole determined in Anaï area from the present study is, on this reference curve (Fig. 8), close to the 270–250 Ma poles (Late Permian) and a little bit farther from 140 Ma pole (Early Cretaceous). An Early Cretaceous age seems however not to be reliable because, for a primary magnetization, such age (that should imply a hiatus over the Paleozoic formations at least of the order of 160 My) should correspond to the Taouratine formation (de Lapparent et al., 1958) unknown in this area (Freulon, 1964). A Late Permian age is in agreement with the only reversed obtained polarity in this study, the main part of the Upper Permian age belonging to the Kiaman superchron of only reversed polarity, but the studied levels could also correspond by a mere chance to reversed polarity periods known during the post-Kiaman times (e.g. Steiner, 2006; Szurlies et al., 2012). The paleomagnetic dating therefore surprisingly indicates that the studied levels at Anaï do not have a Middle–Upper Triassic age. For comparison, the paleomagnetic dating at La Reculée gives an age around 210–170 Ma, i.e. actually Upper Triassic–Lower Jurassic (Fig. 8).

Owing to the general structure of this area, an age around the Permo-Triassic boundary is also very probably in the equivalent levels in the neighboring series to the north in Libya (where they then could perhaps represent the single Permian outcrop) and to the south-east in Niger. Because of the lack of outcrops, we cannot determine if the levels overlying the sampled sites are of Middle–Upper Triassic age or directly the Cretaceous formation. For the

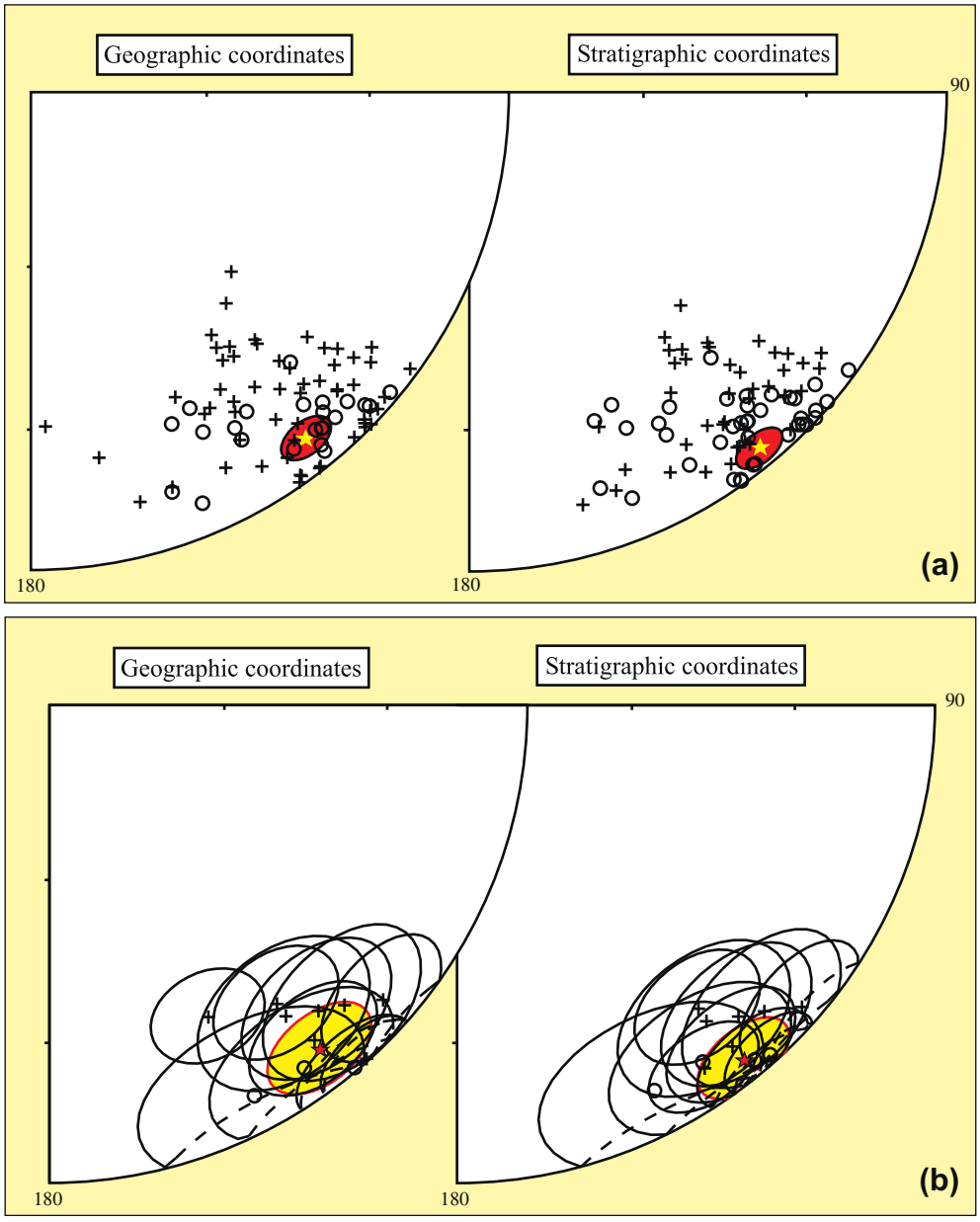


Fig. 7. Equal-area plot (a) of ChRM C, before and after dip correction (crosses: positive inclinations) and (b) of the mean-site paleomagnetic directions with associated 95% confidence zone (continuous line: upper hemisphere; dotted line: lower hemisphere), in geographic and stratigraphic coordinates.

Table 1
ChRM C and great circles data: mean directions obtained for each site (*N*: number of ChRM; *n* number of great circles; *D*: declination; *I*: inclination); paleomagnetic pole: Lat. (°S), Long (°E); and corresponding Fisher parameters (k, α_{95}, K and A_{95}).

Site	*N*	Before dip correction				After dip correction				VGPs and **Paleopole**			
		D (°)	*I* (°)	*k*	α_{95} (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)	Lat. (°S)	Long. (°E)	*K*	A_{95} (°)
7	9	154.5	27.6	36	8.7	147.2	8.9	43	8.0	47.1	64.0	–	–
8	8	143.2	11.6	34	9.7	142.6	9.1	34	9.7	43.6	68.2	–	–
9	4	144.3	19.3	48	13.4	143.2	16.8	48	13.4	41.4	63.6	–	–
12	9	146.4	−8.0	16	13.4	147.0	−10.3	16	13.4	53.0	75.9	–	–
13	5	153.8	−8.5	24	15.8	154.4	−10.3	24	15.8	58.9	67.9	–	–
14	7	137.0	13.0	24	12.6	136.4	10.1	24	12.6	38.4	72.8	–	–
15	12	139.6	0.7	44	6.6	139.7	−2.0	44	6.6	44.6	76.9	–	–
16	5	133.0	7.6	63	9.7	132.7	4.6	63	9.7	37.0	78.3	–	–
18	11	141.4	−1.0	32	8.2	141.5	−3.5	32	8.2	46.6	76.3	–	–
19	7	140.2	15.9	27	11.8	139.4	13.2	27	11.8	39.8	68.8	–	–
20	5	144.2	22.4	61	9.8	143.0	20.0	61	9.8	40.0	62.2	–	–
Mean	82	143.2	8.1	19	3.7	142.3	3.7	22	3.4	–	–	–	–
Mean	11 Sites	143.3	9.2	35	7.8	142.4	5.2	47	6.7	**44.7**	**70.4**	**107**	**4.1**

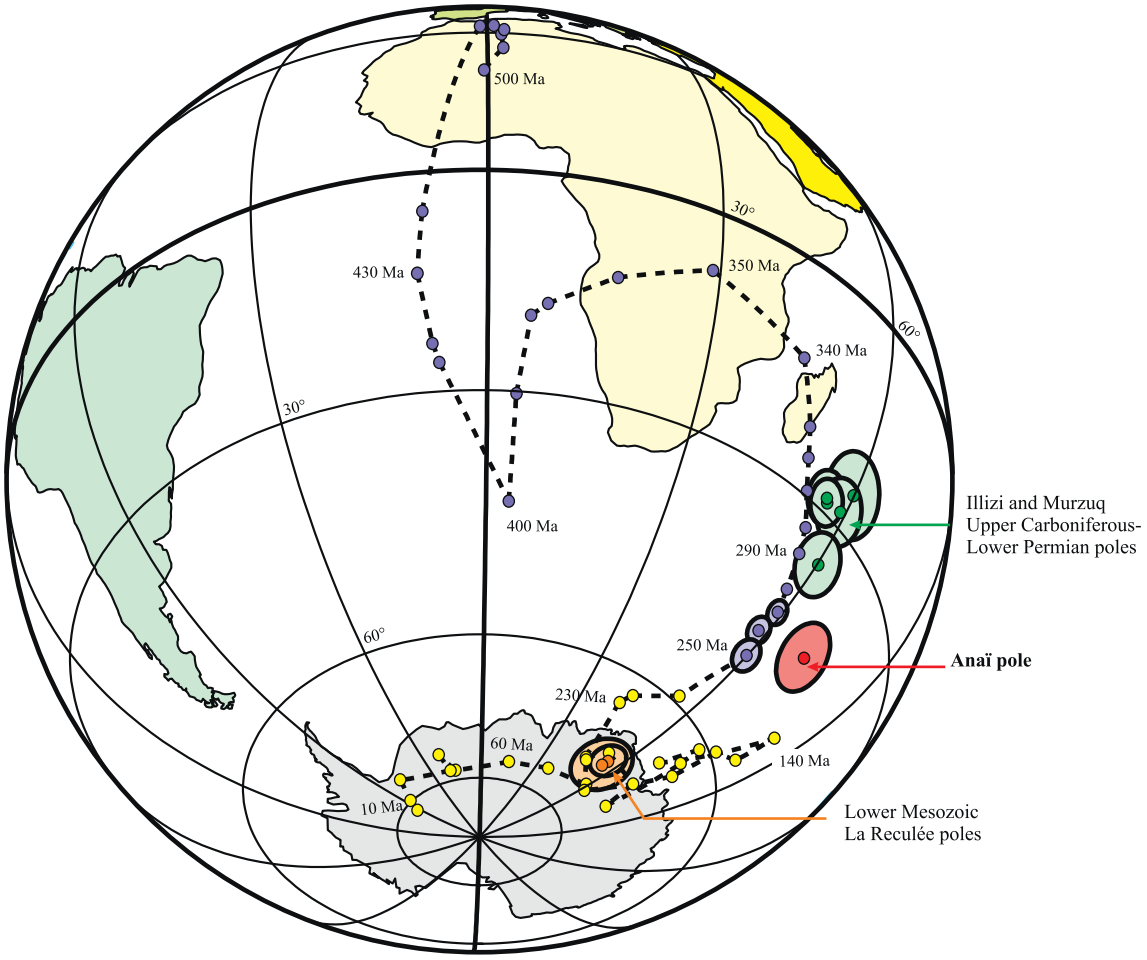


Fig. 8. Comparison of the Anaï pole with the Upper Carboniferous to Lower Jurassic poles of the Illizi basin (see Table 2) and with the Gondwana APWP (500–250 Ma – Amenna et al., 2014 – 240–210 Ma – Domeier et al., 2012 – and 200–0 Ma – Besse and Courtillot, 2002).

Table 2
Paleomagnetic poles for formations of Upper Carboniferous to Lower Jurassic age in the Illizi and Murzuq basins.

Age	Location	Λ (°S)	Φ (°E)	K	A_{95} (°)	Reference
Moscovian	Edjeleh, Algeria	28.3	58.9	157	4.2	Derder et al. (2001c)
Moscovian	In Ezzane, Algeria	25.2	59.9	55.4	5.4	Amenna et al. (2014)
Moscovian	El Abed Larache, Algeria	28.7	55.8	235	2.9	Henry et al. (1992)
Bashkirian	El Abed Larache, Algeria	28.2	55.5	207	3.4	Derder et al. (2001b)
Stephano-Autunian	Tiguentourine, Algeria	35.3	60.3	172	4.0	Derder et al. (1994)
Triassic	La Reculée, Algeria	70.9	55.2	478	2.3	Kies et al. (1995)
Lower Jurassic	Zarzaïtine, Algeria	71.8	54.9	91	3.9	Derder et al. (2001a)

same outcrop reasons, the formation overlain by the studied levels could be the Moscovian (relatively to the Anaï pass, closest outcrops 30 km northward and 70 km toward SSE) or an older formation (Fig. 1).

5.4. Structural and stratigraphic implications

Cartographic discordances are significant on the southwestern border of the Murzuq basin (Fig. 1). The Cretaceous "Continental intercalaire" is known everywhere in this area. For the underlying formations, about 140–145 km and 190–215 km SSE of the Anaï pass (Fig. 1d), the Zarzaïtine formation directly overlies the Lower Carboniferous. That likely highlights an effect of the Hercynian tectonics with the formation of a large anticline with weak curvature. Dips acquired by the series in the anticline are very weak, but presence of such an anticline could be key information for petroleum research. From 145 to 190 km SSE of the Anaï pass (Fig. 1d), the Zarzaïtine formation themselves is lacking (Cretaceous directly overlying the Lower Carboniferous). That can be explained by a lack of deposition on a relatively high elevation part as well as erosion after an uplift. For this last case, that should imply a Mesozoic event similar to that already pointed out on basins of northwestern border of the Hoggar (Smith et al., 2006; Derder et al., 2009).

6. Conclusion

The age difference between Anaï and La Reculée for the Zarzaïtine Formation is considerable, of the order of at least 40 My. The apparent continuity of the outcrops between Anaï and La Reculée areas (Fig. 1) however argues for the belonging to a same

geological formation. The strong diachronism evidenced by the paleomagnetic data for the Zarzaïtine, as well as the local character of the Tiguentourine deposits in newly-formed basins in Libya, clearly indicate that the post-Hercynian evolution of the Saharan platform was not so "quiet". At least vertical movements gave differential uplifts, at the origin of erosion, hiatus or sediments deposition according to areas and very likely according to time for a same area. Such results were not accessible using the classical paleontological and geochronological approaches.

That therefore underlines the original possibilities of the paleomagnetic dating. The latter remains however an under-used method, except for archeological applications and the magnetostratigraphy. It can be based on primary or secondary magnetizations. Its principal limitation is that it needs a well-established reference curve. Such curves are now available for the main continents and can be even used for area having undergone a known local rotation. One of the interests of such a method is dating of geological formations (e.g. Ricordel-Prognon et al., 2010) or of events like impact craters (e.g. Eitel et al., 2014), ore deposits (e.g. Symons and Sangster, 1994), or strong weathering (e.g. Essalhi et al., 2011). But the obtained results could also have important geodynamic implications, for example to date and quantify regional rotation (e.g. Rouvier et al., 2012), deformations (e.g. Lamali et al., 2013) or significant differential vertical movements (this study).

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